

Your Project Title

Project Report

Author(s): Name 1, First Name 1; Name 2, First Name 2, ...



Abstract

This is the abstract. **(150-250 words)** Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Institute for Machine Tools and Industrial Management (iwb)

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L^AT_EX-Tutorial

This tutorial provides a brief introduction to the use of L^AT_EX.

Citation

The standard commands `\textcite` and `\parencite` are available for citations. If the author's name should be used in the sentence, the former is suitable, e.g., Bayerlein et al. [1, pp. 2–3] refers to Bayerlein et al. [2]. If the citation should be placed in parentheses after the statement, the latter is recommended [3]. Grouped citations at the end of a sentence are written as follows: [4, 5]. Citing online sources can be challenging since the author and publication year are not always available. Comparing Heuss and Roder [6] and *iwb-Startseite* [7], it becomes clear that in the latter case, the page title is used instead of the standard schema.

Standards are cited as *DIN ISO EN 10218-2* [8]. Certain settings must be configured in the BibTeX export of the reference management software. The document type should be book with the following entries:

- Standard type and number as title
- Full title as subtitle
- Publisher as publisher
- Year as date
- author must not be assigned!

Abbreviations

Abbreviations can be defined in `./src/abbreviations.tex`. There are specific cases for expressions whose plural forms do not end with "s." In such cases, short and long forms of the expression must also be defined for the plural.

- `\gls{ros}` writes the full form on first occurrence in the document Robot Operating System (ROS), and the abbreviation from the second occurrence onward ROS
- `\glspl{ap}` uses the plural in long form Arbeitspakete (APs) and then in short form AP

Figures

Graphics and images can be included in any file format, as shown in fig. 1. Vector graphics are generally superior to raster graphics in terms of sharpness and storage requirements.

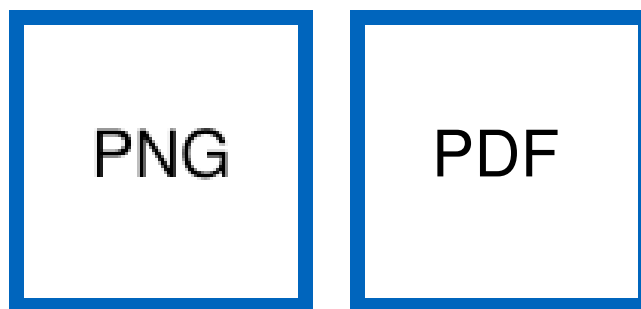


Figure 1: Description of the image.

Table 1: An example table with a column of aligned decimal numbers (S) and a column spanning the remaining width (X).

Left	Aligned	Example text	Right
1	1,110.1	This is quite a long sentence showing that the line wraps in the table.	a
2	10.1	This is quite a long sentence showing that the line wraps in the table.	b
3	23.113,231	This is quite a long sentence showing that the line wraps in the table.	c

1 Introduction (e.g., All 20 %)

This is the introduction.

2 Methods (e.g., All 20 %)

This is the methods.

3 Results (e.g., All 20 %)

This is the results.

4 Discussion (e.g., All 20 %)

This is the discussion.

5 Summary (e.g., All 20 %)

This is the summary.

A Appendix

This is the Appendix.

A.1 Author Contributions

Table 2: Overview of the authors contribution.

Author	Sections	Contribution
Name 1, First Name 1	Introduction	20 %
	Methods	20 %
	Results	20 %
	Discussion	20 %
	Summary	20 %
Name 2, First Name 2	Introduction	20 %
	Methods	20 %
	Results	20 %
	Discussion	20 %
	Summary	20 %

A.2 Declaration of the Use of AI-based Tools

Table 3: Overview of AI-based tools used in the work.

AI-based Tool	Last Date of Access	Use Case	Scope	Remarks
ChatGPT 4.0	04.12.2024	Creating research questions	Ch. 3, pp. 8–11	Chat history in Appendix 2
ChatGPT 4.0	04.12.2024	Code generation for data visualization	Ch. 5, p. 56	Python code
Gemini Pro	18.03.2025	Text revision	Entire thesis	
Microsoft Copilot	09.02.2025	Code generation	Ch. 4, pp. 37–41	Python code, description in Appendix 2
DALL-E 3	09.02.2025	Visualizations	Fig. 5, p. 13; Fig. 7, p. 17	
DeepL Pro	24.11.2024	Translation of text passages	Entire thesis	From Greek to English
Grammarly	17.01.2025	Correction of grammatical and stylistic errors	Entire thesis	Premium subscription

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